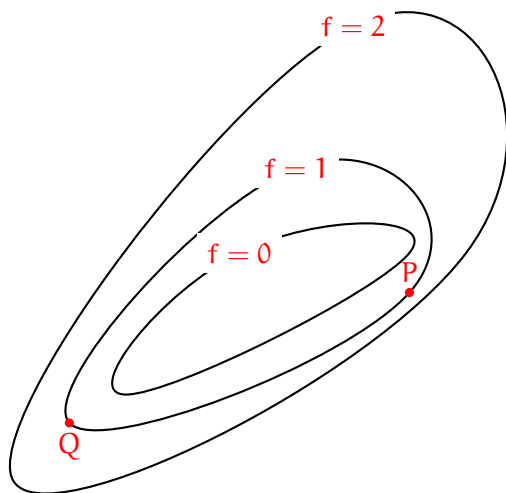


A. **Gradients and Level Sets.** Consider the contour diagram for function f .

We will sketch the gradient vector at various points.



First let us concentrate on the point P .

Starting at P , the direction in which values of f increases most rapidly is

On the other hand, last time we showed that this direction, and its length, are

As the level sets are close at P , the rate of change of f in the gradient direction is

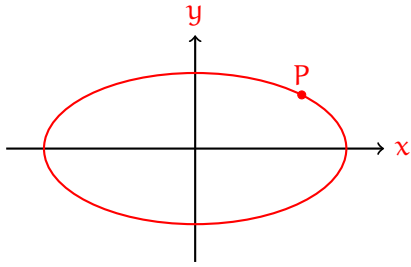
We produce our sketch.

Now we address the point Q .

As before, the gradient points

As the level sets are more distant at Q than at P , the gradient at Q will be

Example 1. Consider the ellipse: $x^2 + 4y^2 = 8$.



(a) Use gradients to parametrize the **normal line** to the ellipse at $P(2, 1)$.

Solution. The normal line is orthogonal to the curve.

We interpret the ellipse as a contour by selecting:

and observing that the ellipse is the contour:

A vector that is orthogonal to the contour at $P(2, 1)$ is

In our case we obtain a normal vector by calculating

$$\vec{\nabla} f(x, y) =$$

$$\vec{n} = \vec{\nabla} f(2, 1) =$$

The normal line is parametrized by

$$\vec{r}(t) =$$

(b) Use the gradient to find an equation for the tangent line to the ellipse at $P(2, 1)$.

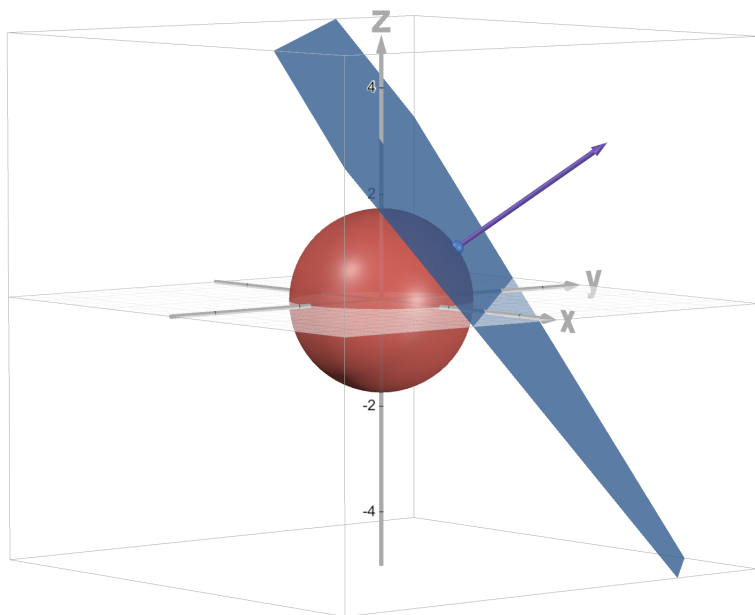
Solution. We have a normal vector to the tangent line:

Next we utilize the equation

which, in two-dimensions, will describe a line given a normal and a point.

In our case

B. Gradients and Tangent Planes. We revisit tangent planes.



Let $F(x, y, z)$ be a differentiable function of three variables.

Let C be a constant and consider the level set:

which, in our case, is actually a level:

Select a point P on this surface.

A vector that points orthogonally to the level surface at P is

$\vec{n} =$

This vector also points orthogonally to the tangent plane.

Let $F(x, y, z)$ be a differentiable function of three variables.

Let C be a constant and P be a point on the level surface $F = C$.

The tangent plane to the level surface at P is defined by

Example 2. Consider the surface defined by $xyz = 8$.

(a) Find an equation for the tangent plane to this surface at $P(2, 2, 2)$.

Solution. First we define:

So that we may interpret the surface as the level surface:

A normal to the level surface is given by the gradient.

In our case

$$\vec{n}(x, y, z) = \vec{\nabla}F(x, y, z) =$$

$$\vec{n}(2, 2, 2) =$$

An equation for the tangent plane is then $\vec{n} \cdot \vec{x} = \vec{n} \cdot \vec{p}$.

(b) Find all points on this surface where the tangent plane is parallel to

$$\mathcal{P} : 4x + 2y + z = 100$$

Solution. A normal to the given plane is

$$\vec{n}_0 =$$

A normal to the tangent plane of the surface at point (x, y, z) is

$$\vec{n}(x, y, z) = \vec{\nabla}F(x, y, z) =$$

For the planes to be parallel, the normals must be parallel. Thus we set up

We also mustn't forget that (x, y, z) need be on the surface, so also require

We put this together to obtain, and solve, the system

Thus the point we are looking for is: